

Yen-Ta Lin

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EDUCATION

Carnegie Mellon University

Master of Science in Electrical and Computer Engineering

Pittsburgh, PA

Expected May 2028

Current Coursework: Large Language Model Systems, Foundations of Computer Systems, Embedded Real-Time Systems

National Cheng Kung University

Bachelor of Science in Electrical Engineering, **GPA:** 4.01/4.30

Tainan, Taiwan

June 2025

Selected Coursework: Systems Programming, Computer Organization, Advanced AI Computing System Design

SKILLS

Programming Languages: Python, C++, C

Machine Learning: PyTorch, Model Quantization, Computer Vision

Embedded & Systems: Linux, STM32, CAN, Oscilloscope, Raspberry Pi

Tools & Infrastructure: Git, Airflow, Kubernetes, CUDA

PROJECTS

National Cheng Kung University

Tainan, Taiwan

LLM Quantization for Edge Deployment - Research Project

February 2025 - June 2025

- Implemented two-scale piecewise INT4 quantization with MSE-calibrated thresholds for Llama-3.2-1B, reducing MLP down-projection component-ablation perplexity by 86.9% versus naive static INT4
- Profiled activation distributions and module-level INT4 sensitivity in Llama models, identifying MLP down-projection activations as a major quantization bottleneck and guiding outlier-mitigation experiments

UAV Search-and-Rescue Vision System - Research Project

September 2024 - January 2025

- Developed an RT-DETR-based UAV person detector on curated aerial search-and-rescue imagery, with 78% precision and 92% recall on fire-department data
- Integrated RT-DETR into a local Streamlit inference application with model loading and execution for field use at Tainan City Government Fire Bureau; project won 1st Place in Tainan City Hackathon and Undergraduate Project Competition

Ki-67 Breast Cancer Cell Segmentation - Research Project

February 2024 - June 2024

- Curated and labeled Ki-67 pathology images and reproduced UniMatch for semi-supervised cell segmentation, establishing the strongest baseline across study experiments
- Applied watershed post-processing to separate adjacent nuclei, reaching 79% mIoU on study data

National Cheng Kung University Formula Racing Team

Tainan, Taiwan

Formula Racing Embedded Control System - Club Project

March 2022 - August 2023

- Programmed STM32F407 firmware and configured CAN communication across six vehicle subsystems, including EMRAX-208 motor control and vehicle integration
- Debugged a CAN bus communication failure using an oscilloscope, traced root cause to an open circuit in an adapter cable, and restored CAN communication to EMRAX-208 motor controller
- Led the Electrical & Controls team for Formula Student Taiwan, assigning technical responsibilities and coordinating subsystem integration; earned 1st place in 2022 and 3rd place in 2023

PROFESSIONAL EXPERIENCE

Mobagel

Taipei, Taiwan

Research and Development Intern (Part-time)

July 2025 - September 2025

- Refactored legacy monitoring code, added unit tests, and rebuilt an Airflow monitoring dashboard in Shiny to replace Google Cloud dashboard

Data Science Intern (Part-time)

February 2025 - June 2025

- Proposed and prototyped TimesNet for time-series forecasting, outperforming existing ML baselines on selected internal datasets
- Integrated an NVIDIA GPU node into Kubernetes to enable GPU workloads; benchmarked runtime performance and ruled out production deployment due to insufficient speedup